Rationalisation

Read the game, then play the odds

In pairs
Two games, six rounds each. Read them.

→

Against the die
Three mixes, six rounds each.

→

Compare
Which reply was best, and why.

In pairs, one of you plays the rows; against the die, you play the **column**.

$$A = \begin{pmatrix} 2 & -2 \\ -1 & 1 \end{pmatrix} \qquad B = \begin{pmatrix} -2 & 2 \\ 1 & -1 \end{pmatrix}$$

A pays the row player, B pays the column player. Rows are r_1 and r_2; columns are c_1 and c_2.

Every cell adds to [ ], so this is a _____ game.

If the row player picks r_1 and you pick c_2, you score [ ] and they score [ ].

While you play

Decide who is the row player and keep that role. Six rounds, both choosing at the same time, no signalling. Fill in every column.

The game: $A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$ $B = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$ rows r_1, r_2; columns c_1, c_2; A pays the row player, B the column player.

Round	a_1	a_2	u_1	u_2	$\sum u_1$	$\sum u_2$
1						
2						
3						
4						
5						
6						

Could either of you have done better by always playing the same action?

While you play

Same pairs, same roles, six more rounds. The payoffs no longer mirror, so read your own cell every time.

The game: $A = \begin{pmatrix} 2 & -2 \\ -1 & 1 \end{pmatrix}$ $B = \begin{pmatrix} -2 & 2 \\ 1 & -1 \end{pmatrix}$ rows r_1, r_2; columns c_1, c_2; A pays the row player, B the column player.

Round	a_1	a_2	u_1	u_2	$\sum u_1$	$\sum u_2$
1						
2						
3						
4						
5						
6						

Did either of you settle on one action? Did it keep working?

While you play

Say c_1 or c_2 out loud before each roll. Six rounds per mix, one D12 covering all three.

The game: $A = \begin{pmatrix} 2 & -2 \\ -1 & 1 \end{pmatrix}$ $B = \begin{pmatrix} -2 & 2 \\ 1 & -1 \end{pmatrix}$ rows r_1, r_2; columns c_1, c_2; A pays the row player, B the column player.

	$\sigma_1 = (1/4,\ 3/4)$ r_1 on 1 to 3			$\sigma_1 = (3/4,\ 1/4)$ r_1 on 1 to 9			$\sigma_1 = (1/3,\ 2/3)$ r_1 on 1 to 4		
Round	a_2	**die**	$\sum u_2$	a_2	**die**	$\sum u_2$	a_2	**die**	$\sum u_2$
1									
2									
3									
4									
5									
6									

Totals after six rounds:

Against $\sigma_1 = (p,\ 1-p)$ your two actions pay

$u_2(c_1, \sigma_1) =$ $u_2(c_2, \sigma_1) =$

So c_1 is the better reply exactly when p , and the two are equal at $p =$.

Is c_1 dominated by c_2?

Is c_2 dominated by c_1?

Neither, because which one is better depends on _____.

Dominance only settles an action that is worse against *every* strategy of the opponent. Here each action is a best response to some mixes, so we need the _____ instead.

Against the third die both of your actions pay [ ], so every strategy of yours is a best response and your six-round total should be about [ ].

That is the row player's half of the equilibrium: they roll precisely so that you cannot exploit them. Which of your own mixes does the same to *them*?

Dominance and best responses are not examined on their own. They are examined together with **Nash Equilibrium**, so the natural follow-up is **Question 1** on that page, where the best response condition does exactly this job for Rock-Paper-Scissors.

The habit worth building now: before computing anything, ask which actions could ever be a best response. Anything that never can be is not worth carrying into the algebra.

vknight.org/gt/topics/nash-equilibrium.html

Every stage of today, and the part of the notes it was reading.

What we did	Notes	In the textbook
Playing against a known mix	**2.2.4**	Games: Definition: Strategy in Normal Form Games
Looking for a dominated action	**3.2.2, 3.2.3**	Rationalisation: Strictly Dominated, Weakly Dominated Strategy
Choosing against the die	**3.2.5**	Rationalisation: Definition: Best Response
The mix that left you indifferent	**3.2.6**	Rationalisation: Theorem: Best Response Condition
Examinable	**3.2.6**	Theorem: Best Response Condition, and proof

vknight.org/gtb/chapters/games, vknight.org/gtb/chapters/rationalisation

Quick quiz before you go: vknight.org/gt/topics/nash-equilibrium.html

Before the next class: read the Rationalisation chapter as far as the best response condition, and attempt Question 1 on the Nash Equilibrium page.